Secure Monitoring Platform Requirements

Software Requirements Specification

IEEE 830-1998 outline

DRAFT — human review required

Version 1 | 2026-10-29T20:07:17.239753+00:00

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1 Introduction

1.1 Purpose

This specification records the project purpose, stakeholder constraints and unresolved decisions stated in the source evidence below.

This SRS describes the software's required behavior for users, administrators and the Project Analyst reviewing the specification.

1.2 Scope

Design a secure monitoring platform for university servers and managed endpoints that collects system events, authentication activity, process information, and security alerts from multiple departments. The platform should detect unusual login behavior, repeated failed authentication, suspicious process activity, and unauthorized configuration changes. Security administrators should be able to review alerts, investigate affected devices, record investigation notes, change alert status, and isolate compromised endpoints when necessary. The system should maintain audit logs of security events and administrator actions and provide a centralized dashboard for monitoring security status across multiple campus locations.

1.3 Definitions, acronyms and abbreviations

Term	Meaning
Device isolation	Restriction of a device's network access; the source determines whether and by whom this action is permitted.
Log	A collection of event records. This term does not itself require additional logging fields, events or access permissions.
Audit record	A stored record of an event. Required event scope and retention are determined by the linked source, not this definition.

1.4 References

Reference	Source
Format	IEEE Std 830-1998, clauses 5 and Annex A (document organization).

1.5 Overview

Section 2 summarizes the product, stakeholders and open decisions. Section 3 contains the uniquely identified requirements. Detailed evidence, acceptance procedures and the SDLC plan are available in the supporting report.

2 Overall description

2.1 Product perspective

Design a secure monitoring platform for university servers and managed endpoints that collects system events, authentication activity, process information, and security alerts from multiple departments.

2.2 Product functions

Functional topics are summarized below; section 3 retains the full requirements.

- collect system events, authentication activity, process information, and security alerts from multiple departments for university servers and managed endpoints - FR-001
- detect unusual login behavior - FR-002
- detect repeated failed authentication - FR-003
- detect suspicious process activity - FR-004
- detect unauthorized configuration changes - FR-005
- allow Security administrators to review alerts - FR-006
- allow Security administrators to investigate affected devices - FR-007
- allow Security administrators to record investigation notes - FR-008
- allow Security administrators to change alert status - FR-009
- allow Security administrators to isolate compromised endpoints when necessary - FR-010
- maintain audit logs of security events and administrator actions - FR-011
- provide a centralized dashboard for monitoring security status across multiple campus locations - FR-012

2.3 User characteristics and stakeholders

Labels group stakeholders for readability. They do not combine permissions.

Education, training and experience requirements remain unspecified.

Role	Source user groups	Responsibility
Admin	administrators	Performs only the role's source-defined operations in section 3.
Project Analyst	Document review role	Reviews requirements and tracks open decisions; this role grants no product access.

2.4 Constraints

No additional implementation constraints were confirmed. Source-defined restrictions remain in the requirements in section 3.

2.5 Assumptions and dependencies

Open decisions requiring confirmation:

- What conditions define unusual login behavior?
- What conditions define repeated authentication failures?
- What conditions define suspicious process activity?
- What makes a configuration change unauthorized?
- What conditions authorize endpoint isolation?
- Which fields must the audit logs contain?
- What retention period is required for the audit logs?

- Which security acceptance outcomes must be demonstrated?
- How are the stated roles and authorized permissions assigned?

3 Specific requirements

Each requirement is stated once with its original identifier and wording. Cross-references identify requirements covering more than one concern.

3.1 External interface requirements

3.1.1 User interfaces

Related requirements: FR-012. Interface details beyond those requirements remain unspecified.

3.1.2 Hardware interfaces

Related requirements: FR-007. Interface details beyond those requirements remain unspecified.

3.1.3 Software interfaces

Related requirements: FR-001, FR-003. Interface details beyond those requirements remain unspecified.

3.1.4 Communications interfaces

No specific interface requirement was supplied; confirmation is required.

3.2 Functional requirements

FR-001 · Must · explicit

The system shall collect system events, authentication activity, process information, and security alerts from multiple departments for university servers and managed endpoints.

Traceability: E-001, TEST-001

FR-002 · Should · explicit

The system shall detect unusual login behavior.

Traceability: E-002, TEST-002

FR-003 · Should · explicit

The system shall detect repeated failed authentication.

Traceability: E-002, TEST-003

FR-004 · Should · explicit

The system shall detect suspicious process activity.

Traceability: E-002, TEST-004

FR-005 · Should · explicit

The system shall detect unauthorized configuration changes.

Traceability: E-002, TEST-005

FR-006 · Should · explicit

The system shall allow Security administrators to review alerts.

Traceability: E-003, TEST-006

FR-007 · Should · explicit

The system shall allow Security administrators to investigate affected devices.

Traceability: E-003, TEST-007

FR-008 · Should · explicit

The system shall allow Security administrators to record investigation notes.

Traceability: E-003, TEST-008

FR-009 · Should · explicit

The system shall allow Security administrators to change alert status.

Traceability: E-003, TEST-009

FR-010 · Should · explicit

The system shall allow Security administrators to isolate compromised endpoints when necessary.

Traceability: E-003, TEST-010

FR-011 · Should · explicit

The system shall maintain audit logs of security events and administrator actions.

Traceability: E-004, TEST-011

FR-012 · Should · explicit

The system shall provide a centralized dashboard for monitoring security status across multiple campus locations.

Traceability: E-004, TEST-012

3.3 Performance requirements

No separate requirement was specified in the supplied brief.

3.4 Logical database requirements

No separate requirement was specified in the supplied brief.

3.5 Design constraints

No separate requirement was specified in the supplied brief.

3.6 Software system attributes

3.6.1 Reliability

No separate requirement was specified in the supplied brief.

3.6.2 Availability

No separate requirement was specified in the supplied brief.

3.6.3 Security

Also covered by: FR-001, FR-003, FR-006, FR-007, FR-008, FR-009, FR-010, FR-011, FR-012.

NFR-001 · Must · explicit

The system shall be secure.

Traceability: E-001, TEST-013

3.6.4 Maintainability

No separate requirement was specified in the supplied brief.

3.6.5 Portability

No separate requirement was specified in the supplied brief.

3.7 Other requirements

No separate requirement was specified in the supplied brief.

Detailed acceptance criteria, evidence, architecture and the SDLC plan are available in the supporting report and canonical JSON.

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